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Decision curve analysis: a novel method for evaluating prediction models

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Role in this programme

Origin of decision curve analysis. Chapter 3 lists DCA among the evaluation methods for judging whether a model is clinically useful, not merely discriminative, which matters because accuracy is close to uninformative under this severity distribution.

ABSTRACT

Diagnostic and prognostic models are typically evaluated with measures of accuracy that do not address clinical consequences. Decision-analytic techniques allow assessment of clinical outcomes, but often require collection of additional information, and may be cumbersome to apply to models that yield a continuous result. The authors sought a method for evaluating and comparing prediction models that incorporates clinical consequences, requires only the data set on which the models are tested, and can be applied to models that have either continuous or dichotomous results. The authors describe decision curve analysis, a simple, novel method of evaluating predictive models. They start by assuming that the threshold probability of a disease or event at which a patient would opt for treatment is informative of how the patient weighs the relative harms of a false-positive and a false-negative prediction. This theoretical relationship is then used to derive the net benefit of the model across different threshold probabilities. Plotting net benefit against threshold probability yields the decision curve. Decision curve analysis identifies the range of threshold probabilities in which a model is of value, the magnitude of benefit, and which of several models is optimal.